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MASTERCARD DATA QUEST · CASE CHAMPIONSHIP

Detecting Hidden Commercial Activity in Retail Transaction Data

An explainable ML model that flags self-employed “hidden entrepreneurs” who run a business on ordinary consumer cards — turning transaction behaviour into B2B sales leads.

Team Seamless

23 May 2026

BUSINESS CONTEXT

The opportunity: businesses hiding inside the consumer base

A segment of sole traders and micro-businesses runs commercial activity through ordinary **consumer** cards.

Their behaviour already looks like a business: frequent POS purchases, recurring platform charges, large tickets and a narrow set of merchant categories.

OUR OBJECTIVE



Build a predictive model that identifies hidden entrepreneurs from transaction behaviour — reliably, and with explainable reasoning.



Acquiring & transaction revenue

Business cards carry different tariffs and unlock POS-acquiring and merchant services — a new fee stream.



Cross-sell

Working-capital lending, payroll projects, trade acquiring and cashier solutions for the right clients.



Precision targeting

Stop spending outreach budget on ordinary consumers — call only the clients who actually need B2B products.

THE DATA

Two labelled card populations + a merchant reference

25K

Business cards

~3.0M transactions

80K

Consumer cards

~9.8M transactions

~2K

Merchants

MCC reference table

LABELLING STRATEGY

Business card → target = 1

Consumer card → target = 0

The model learns the behavioural fingerprint of a known business, then scores consumer cards for the same pattern — the hidden entrepreneurs.

WHAT EACH TRANSACTION TELLS US

amount · MCC category · merchant · channel (online/offline)
country · timestamp · card tier · tokenised wallet · is-recurring

Period: 1 Oct 2025 – 31 Mar 2026 (6 months)

APPROACH

One reproducible pipeline, transaction → decision



Load & label

1

Merge both card files; tag business = 1, consumer = 0



Clean

2

Defensive fills; drop amount outliers (1st–99th pct, ~2%)



Aggregate

3

Collapse 12.8M rows to one row per card



Engineer 14 features

4

Behavioural signals tied to business hypotheses



Train & compare

5

LR baseline · Random Forest (tuned) · Gradient Boosting



Score & segment

6

Probability 0–1 → HIGH / MEDIUM / LOW tiers

Single random seed (42) · stratified split · fixed library versions — every step re-runs end-to-end without manual edits.

DATA PROCESSING

Clean inputs before any feature is built



Missing values

- Defensive fills: merchant fields → "Unknown"
- recurring_capable / is_recurring → False
- This dataset had no missing values — no rows dropped



Outliers in amount

- Rows beyond the 1st / 99th percentile removed
- 256,533 rows (~2%) trimmed as amount outliers
- Protects medians & per-card averages from distortion



Time features

- Parsed timestamp → hour of day
- Derived weekday / weekend flag
- Feeds business-hour & weekend behaviour signals

FEATURE ENGINEERING

14 behavioural features — each answers a business question

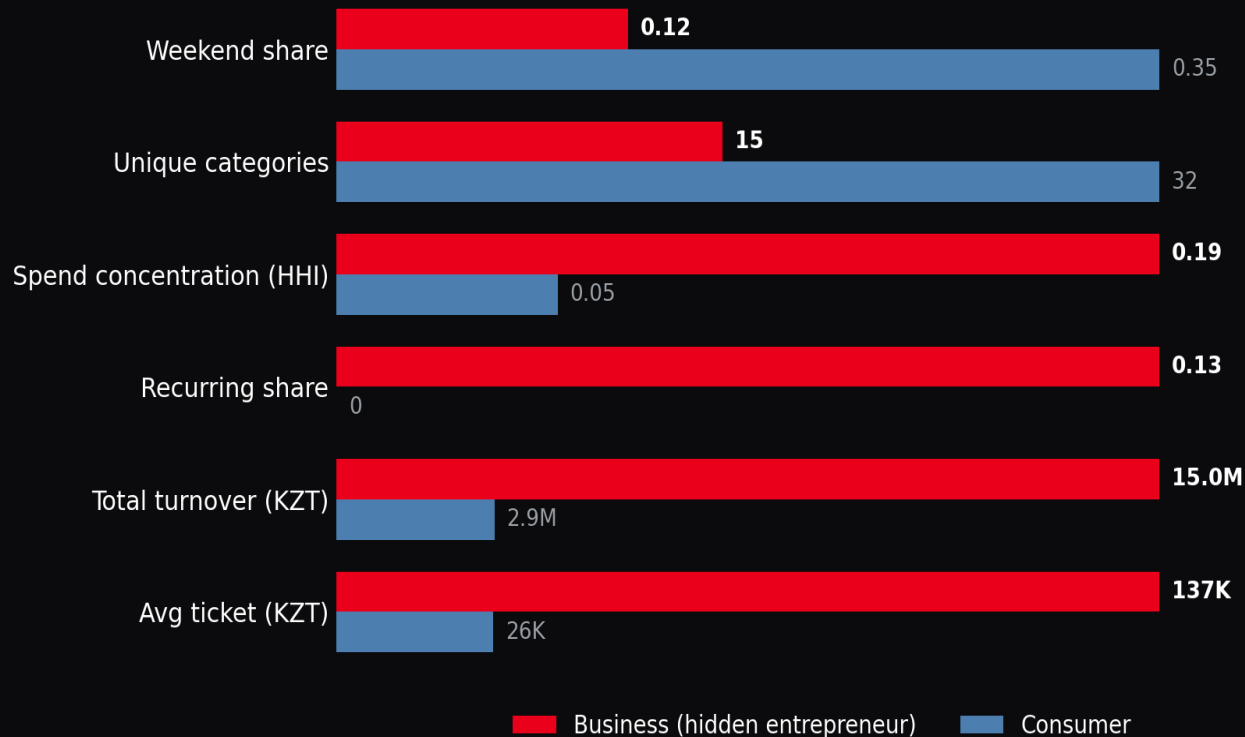
FEATURE(S)	WHAT IT MEASURES	BUSINESS HYPOTHESIS
recurring_ratio	Share of merchant-initiated recurring charges	Cloud, ads, logistics subscriptions = B2B
tx_mean / tx_sum	Average ticket & total turnover	Businesses buy in bulk — larger spend
unique_mcc / mcc_hhi	Category count & Herfindahl concentration	A business focuses on a few categories
unique_merchants	Distinct merchants used	A recurring supplier pool, not random retail
offline_ratio	Share of in-person transactions	Merchant / trade acquiring is offline-heavy
intl_ratio / unique_countries	Foreign-transaction share & geography	Importers and cross-border sellers
business_hour_ratio / weekend_ratio	When spending happens	Activity in work hours, quiet weekends
tx_count / active_days / tx_std	Volume, regularity & spread of spend	Operational rhythm vs household noise

All features are aggregated to the unique-card level, so the model judges a cardholder's whole 6-month behaviour — not single transactions.

EXPLORATORY ANALYSIS

The two groups really do behave differently

Median Behaviour: Business vs Consumer (bars normalised, labels = real medians)



Weekday-driven

Lower weekend share is the clearest tell — business activity follows the working week.

Concentrated spend

Higher MCC concentration (HHI) and fewer categories — focused, not scattered.

More recurring

Recurring charges are a near-zero habit for consumers but routine for businesses.

Bigger tickets

Business median ticket $\approx$ 137K KZT vs 26K — a $\sim 5\times$ gap.

MODELLING

Three models compared — Random Forest wins

MODEL	ROLE	TEST F1	SELECTED
Logistic Regression	Interpretable baseline	0.989	
Random Forest (tuned)	Final model — GridSearchCV, 5-fold	0.997	✓
Gradient Boosting	Close alternative	0.997	

Stratified 80 / 20 split (class balance preserved) and 5-fold GridSearchCV. Random Forest and Gradient Boosting tie on F1 (~0.997); Random Forest is chosen for the highest test F1 plus native interpretability (feature importance + SHAP).

WHY RANDOM FOREST

✓ Built for structured, tabular data

Tree ensembles are the industry standard for client-metric tables — well ahead of linear models here.

✓ Captures feature interactions

A high ticket or a specific MCC alone may mean nothing; their combination is a sharp business marker.

✓ Robust to transaction anomalies

One-off huge purchases don't derail it the way they do simpler models.

✓ Explainable by design

Native feature importances + SHAP satisfy the explainability requirement out of the box.

Model Performance — Random Forest

0.997

F1 Score

Balanced precision & recall

1.000

ROC-AUC

Perfect discrimination

0.9997

PR-AUC

Strong on imbalanced classes

0.996

CV F1 (train)

5-fold stratified validation

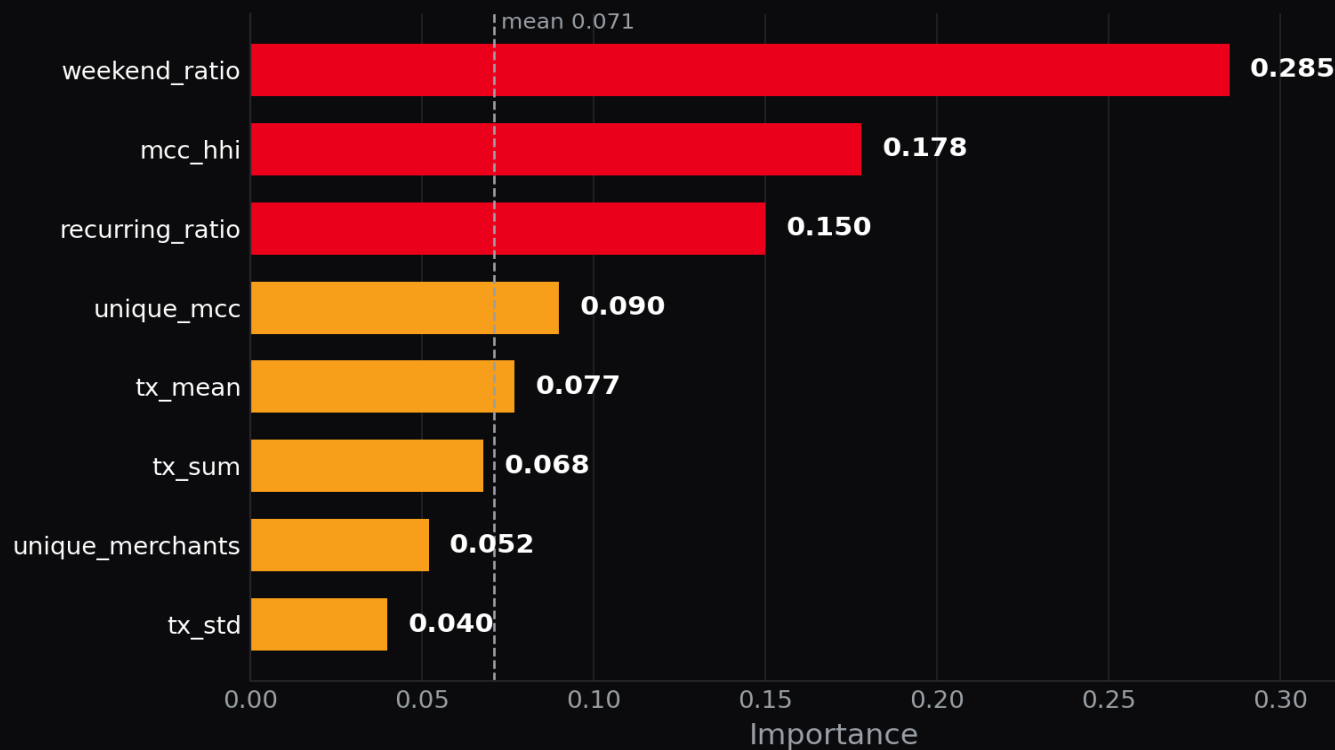
Threshold Analysis — Choose by Business Goal

Threshold	Precision	Recall	F1	Recommended Use Case
0.30	0.994	0.999	0.997	Mass-market outreach — maximum coverage
0.40	0.995	0.999	0.997	Broad targeted campaign
0.50	0.996	0.997	0.997	Sales team prioritization (default)
0.60	0.998	0.995	0.997	Premium cross-sell targeting
0.70	0.998	0.994	0.996	Only highest-confidence cases

EXPLAINABILITY

What drives a “business” verdict

Feature Importances — Random Forest



1

weekend_ratio

The strongest signal. Businesses spend on a work-week rhythm — low weekend activity. Consumers peak on weekends.

2

mcc_hhi

Spend concentration. A business funnels money into a few merchant categories; consumers scatter widely.

3

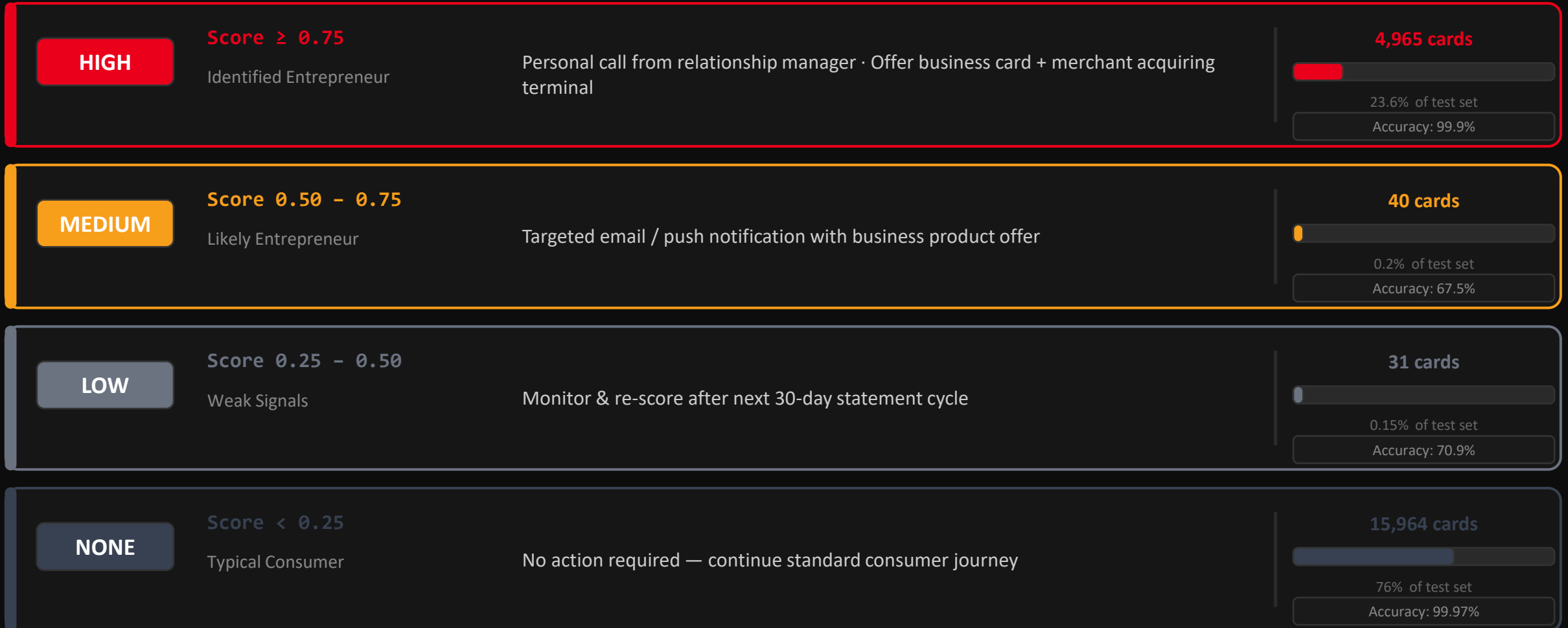
recurring_ratio

Merchant-initiated recurring charges — subscriptions and auto-billing that come with running a business.

Importances cross-checked with SHAP values — direction and ranking agree, so the model’s logic is auditable.

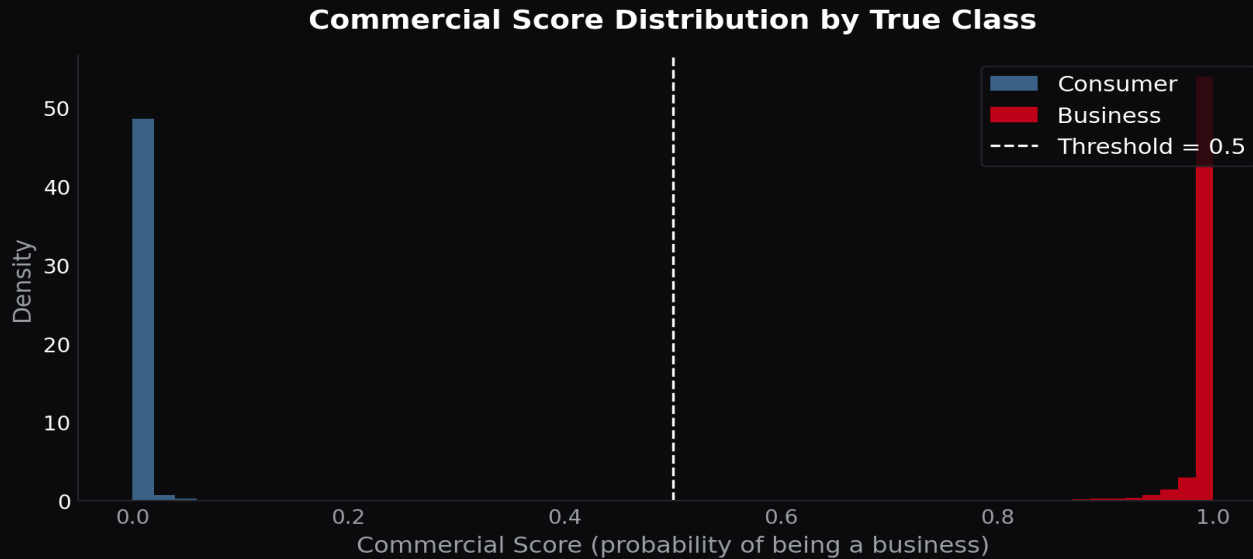
Commercial Scoring & Customer Segmentation

Each card receives a commercial score (0–1) and is assigned to an action tier:



ACTIVATION

From a probability score to a sales action



The model outputs a 0–1 commercial score, so the bank can move the threshold to match each campaign’s goal — wide reach (0.3) or precise premium targeting (0.7). On the test set scores split cleanly: 4,965 HIGH and 15,964 NONE, with only 71 cards in the ambiguous middle.



HIGH ≥ 0.75

Clear hidden entrepreneur · 4,965 cards

Personal manager call — offer business card + acquiring



MEDIUM 0.50–0.75

Probable entrepreneur · 40 cards

Targeted email with B2B product offer



LOW 0.25–0.50

Weak signals · 31 cards

Monitor; re-evaluate in 30 days

LIMITATIONS & NEXT STEPS

Honest caveats and a path to production

Limitations

- Cold start** — Cards with < 2–4 weeks of history score unreliably.
- Synthetic data** — Near-perfect scores (F1 0.997, AUC 1.000) come from highly separable generated data; expect lower numbers on real data.
- Validation gap** — Tested on known business vs consumer cards — detecting hidden businesses inside consumer cards is an unproven extrapolation.
- Pattern adaptation** — Entrepreneurs may split spend across cards, eroding accuracy over time.
- No inbound flows** — Transfers from many counterparties would be a strong signal but aren't in the data.

Recommendations

- Re-train quarterly** — Refresh on recent data to stay ahead of behaviour drift.
- Run an A/B test** — Validate the extrapolation on real clients: HIGH-tier targets vs control on business-card conversion.
- Monitor data drift** — Track PSI on key features to catch degradation early.
- Reproducible by design** — Seed 42, stratified split, pinned libraries — re-runs to the same result.

Business Potential & Revenue Upside

Business Card Upgrades

Impact: HIGH

- Higher annual fee vs consumer card
- Better interchange on B2B MCCs
- Premium tier product attach

Merchant Acquiring

Impact: HIGH

- POS terminal installation
- QR & NFC acquiring fees
- Loyalty program attachment

Working Capital Lending

Impact: MEDIUM

- Business overdraft products
- Short-term revolving credit
- Invoice financing solutions

Payroll & B2B Services

Impact: MEDIUM

- Employee salary cards
- Tax reporting integrations
- ERP / accounting connectors



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Hidden businesses, now visible.

An explainable Random Forest turns six months of card behaviour into a ranked list of B2B sales leads — high recall, clean precision, and a transparent reason behind every flag.

Thank you · Team Seamless · Mastercard Data Quest